CCT College Dublin Continuous Assessment

Programme Title:	HDIP Data Analytics		
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Delivery Mode:	FT/PT		
Cohort Details:	*HDipData_Sept25_ FT Stage 1 Sem1*		
Module Title(s):	*Data Preparation Programme schedules are all published on the CCT IQR Provider Profile*		
Assignment ype:	*Individual*	**Weighting(s):**	*50%*
Assignment Title:	CA1_DataPrep_HDip		
Lecturer(s):	David McQuaid		
Issue Date:	*13/03/2026*		
Submission Deadline Date:	*19/04/2026 at 11:59pm*		
Late Submission Penalty:	Late submissions will be accepted up to **5** calendar days after the deadline. All late submissions are subject to a penalty of **10%** of the mark awarded. Submissions received more than 5 calendar days after the deadline above **will not** be accepted and a mark of 0% will be awarded.		
Method of Submission:	**This assignment is submitted via Moodle.**		
Instructions for Submission:	The Jupyter Notebook File Must be saved as "YourName_DPrepHDip_CA1.ipynb"		
Feedback Method:	**Results posted in Moodle gradebook**		
Feedback Date:	Approx. 3 weeks after FINAL submission (inc PMC cases)		

Introduction

The following report presents the data preparation process performed on a university performance dataset. Our main objective in this phase is to ensure that the dataset is clean, consistent, and suitable for subsequent analysis and modeling. The process includes exploration, cleaning, transformation, visualization, and dimensionality reduction of the data using Principal Component Analysis. Because this work is a university-level learning process, comments were made during coding and annotations were taken where challenges and the need for assistance were identified. I perform the work and separation in parts, where we delve a little deeper into each part to understand every aspect of the process.

Initial Data Exploration and Dataset Characterisation

The dataset was initially loaded and explored using the pandas library. Some basic data inspection methods, such as .head(), .shape(), .info(), and`.describe(), were used to understand its structure and content.

According to the results obtained, our dataset delivered to the project contains 4424 observations and 37 attributes, where each row represents a student and each column corresponds to demographic, academic, or socioeconomic characteristics.

The dataset includes a mix of data types:

- Integer (int64)
- Floating-point (float64)
- Categorical (object)

This combination indicates that different preprocessing techniques are needed depending on the variable type.

The target variable, "Target", contains three categories:

- dropout
- enrolled
- graduate

This confirms that we will be working on a multi-class classification problem.

Handling Missing Values

After analyzing the missing values, it was revealed that only a very small number of entries were missing from our data. For the most part, only one value per column was missing.

Given the size of the dataset (4424 rows), however, the data did not represent large differences, even leaving one case insignificant.

The decision made was that instead of applying imputation techniques, the rows containing missing values were removed, and how did we do this? With the following command:
df = df.dropna()

Resulting in a reduced dataset from 4424 to 4415 rows, only 9 rows removed and no missing values remaining.

This approach provided us with the minimization of data loss, also avoiding the introduction of artificial bias, and lastly, the integrity of the dataset was guaranteed.

Data Cleaning and Type Conversion

During the data inspection process, some inconsistencies were identified, such as the presence of invalid symbols, like '?', and also some numeric values stored as strings.

These inconsistencies were corrected by replacing invalid entries with NaN, as specified in the notebook, and converting the columns to numeric format.

This ensures that all variables are formatted correctly and only the target variable remains categorical.

Target Variable Distribution

A count plot was used to facilitate verification of the distribution of the target variable.

The results show that the majority of students are in the "graduates" category, followed by "dropouts," and lastly, "enrolled" is the smallest group. This indicates a moderate imbalance between the classes.

Furthermore, inconsistent labels such as "grad" and "drop" were standardized to:

- graduate
- dropout

Understanding the distribution of classes is crucial because unbalanced data can directly affect the model's performance, and some classes may be underrepresented.

Distribution of Numerical Variables

Histograms were used to analyze the distribution of numerical variables. This part of the project was a bit more challenging, requiring meetings with other students, data analysis, and ensuring we were on the right track.

We were able to make several observations, such as: many variables exhibit an asymmetrical distribution, which is not normal. Some variables are categorical but numerically coded. We also observed the presence of potential outliers. We can also assess that some variables have an approximately normal distribution.

We also had some implications:

The data are not naturally standardized. Scaling is necessary before applying Principal Component Analysis (PCA).

Outlier Analysis

Box plots were used to identify extreme values.

Results obtained in this stage showed that variables such as admission grade and previous qualification grade presented notable outliers.

However, the results for academic performance variables showed consistent distributions.

In this stage, the outliers were retained because:

They represent real-world variations and do not appear to be errors in the data.

Correlation Analysis

A heatmap was generated to analyze the relationships between the numerical variables, where we can analyze some results such as:

Most variables show weak correlations.

There are strong correlations between: Academic performance characteristics and variables of the curricular units.

We can conclude that this indicates local redundancy in some groups of characteristics and also supports the use of PCA for dimensionality reduction.

Feature Scaling Before PCA

Before applying PCA, the target variable was separated from the dataset, as it represents the outcome to be predicted. Only numerical variables were used, since PCA works with numerical data.

Following this, the data were standardized using the StandardScaler. This step ensures that all variables are on the same scale, as there were variables on different scales, preventing variables with larger values from influencing the results more. As a result, the PCA analysis becomes more balanced and reliable.

PCA Results and Interpretation

Principal Component Analysis was applied to reduce dimensionality, resulting in:

- Original features: 36
- Components needed for 99.5% of the variance: 33

This can be interpreted as follows: the reduction is minimal, from 36 to 33, and also indicates low overall redundancy.

Then the correlation analysis showed local redundancy. We can conclude that the dataset presents high informational complexity.

PCA confirms the structure, but does not significantly compress the data.

PCA Visualisation

Finally, a scatter plot was generated using the first two principal components. There was no clear separation between the classes or significant overlap between the categories.

At this stage of the project, we can conclude that the classes are not linearly separable and the dataset is complex, and also that PCA alone is insufficient for classification.

Conclusion

In this project, the main focus was preparing the dataset for analysis. This included exploring the data, cleaning it, correcting errors, and understanding the behavior of the variables.

The dataset was, for the most part, clean, with very few missing values (; and ?). Therefore, removing these rows did not have a significant impact. Some inconsistencies were also corrected, and the data types were adjusted to allow for proper analysis.

From the visualizations, it was possible to observe that many variables do not have a normal distribution, and some contain outliers. However, these outliers appear to be part of the real data and were retained. The target variable also showed some imbalance between classes, which could be a danger for the future.

After applying PCA, the results showed only a small reduction in dimensionality. This suggests that the dataset still contains a high level of information. The PCA scatter plot also showed that the classes are not clearly separated, indicating that the problem is more complex.

Some difficulties and challenges were encountered in the project, requiring us to meet with other students and discuss which results were most consistent.

Overall, this project demonstrates the importance of data prepration. PCA helped to better understand the structure of the dataset, but also showed that more advanced techniques may be necessary in the next steps.

References (Harvard Style)

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